

Integrating into the design process the needs of those involved in the product life-cycle

G. PRUDHOMME^{†*}, P. ZWOLINSKI[†] and D. BRISSAUD[†]

Numerous design methods exist but models and tools used by these design methods do not realize a complete coherence between customers' needs, professionals' needs and solutions that can be found. So, in this paper we will examine the functional analysis method that was specifically developed to allow needs to be taken into account during the design stages. We will present our interpretation of that method and how it could serve to assure this coherence. We will put forward a design activities model, integrating into the design process the needs of those involved in the product life-cycle. Our observations relative to the implementation of the functional analysis method will lead us to supplement this initial model and suggest tools to help designers.

1. Introduction

Companies today are confronted with a problem of competitiveness that requires them to design quality products which meet the needs of the customer, at the lowest possible cost and within the shortest possible lead times. To adapt to this new economic context, they have had to modify their approach to design and adjust their organizational approach. The integrated-design approach, implemented as part of a simultaneous engineering methodology (Solhenius 1992), is increasingly being preferred to sequential-type design models. Integration was initially seen as a transfer of knowledge from the latter stages of the product life-cycle to those involved in the earlier phases of the product life-cycle. This led to efforts being made to capitalize on the knowledge of those working in those areas. It was necessary to make it available to designers in the form of help tools such as, for example, expert computer systems (Descotte and Latombe 1981) or 'guidelines' (Pahl and Beitz 1996). The limitations of such design methods became apparent very quickly: because the knowledge harboured by those professions involved in the design phase is contextual, dynamic and perpetually evolving, it would be futile to hope to integrate it into the design office context in the form of tools. The way the design function was organized subsequently moved towards the creation of project teams. These teams were made up of people from the different professions involved in product development and grouped together in one area to allow them to interact, and indeed encourage them to do so. This means that we are no longer reasoning in terms of single designers, but designers of different technical backgrounds (production, assembly, maintenance, recycling, etc.) and different economic backgrounds (marketing, purchasing, etc.) so that the design context could be taken into consideration (Maffin 1998). Thus, putting the emphasis on design means integrating into the design process not the actual knowledge itself, but the actors who

Revision received January 2003.

[†] Laboratoire Sols Solides Structures, University of Grenoble, BP 53, 38041 Grenoble cedex 9, France.

* To whom correspondence should be addressed. e-mail: guy.prudhomme@hmg.inpg.fr

generate and manipulate that knowledge and help it to evolve (Garro et al. 1996). Initially seen as an issue of integrating knowledge, we have progressed towards considering how to integrate individuals. This requires: 1) that the organization of these companies changes to allow this new form of collaboration to take place (Pawar et al. 1999), leading to a so-called 'concurrent engineering' type organisation; and 2) that tools are made available that allow the different actors of the product life-cycle to interact.

In the present article, we will be discussing tools in particular, and more particularly those that promote the expression and tracking of needs and expectations in relation to the products being designed. Traditionally, the need is the domain of the customer, but the customer is not always the end-user. It could be a buyer whose objective is to modify the practices of those who will use the product under design. According to current design practices, the end-user or the buyer are rarely effective actors in the design phase. They are, nevertheless, represented by one or a number of specific actors: those concerned with marketing or ergonomics, for example. The product is therefore designed for the customer via his/her representative.

This strategy, where the customer's needs are taken into account at the beginning of the design phase, is not the only design model used. In particular, it does not affect projects for which there is no negotiation possible between the customer and the supplier (e.g. the relationship between the buyer and the supplier in the automobile business). It does, however, present a real interest within the framework of projects where a true customer-supplier relationship is established. Typically, in this type of project, the designers work according to the needs of customers as they appear in the initial specification. This includes the unsaid, the implicit and the undefined, supposedly part of the cultural elements that are shared by the members of the project group (manufacturing facilities, maintenance facilities, etc.). Moreover, the expectations may be contradictory because they are optimized according to the needs of different entities (e.g. minimum volume and maximum effort resistance). We will be referring to that type of project in the present paper.

In the sequential model, customer needs appear first of all. The designers working in the design office will transform these customer needs into specifications, which the other actors involved in the product life-cycle will be required to adhere to. The needs that are specific to the professions of those actors (to ensure the product can be made, maintained, stored, recycled, etc.) will be perceived as constraints that have to be taken into consideration during the development of a suitable product for the customer. Taking the needs of these professions into account led to the development of methods such as Design for X (DFM, DFA, DFE, etc.) (Boothroyd et al. 1994). Today, for a simultaneous design model in a concurrent engineering environment, there are two notions concerning the needs: the customer's need, to express the customer's expectations; and the professional's need, to express the expectations of those involved in the product life-cycle. These two notions represent things of the same nature, which have to be explained and defined at the very beginning of the design procedure.

In a project such as this, to integrate the actors who manipulate knowledge rather than of the knowledge itself, we are faced with the following questions:

- What place should be reserved for needs (customers' and professionals') in a design process? What tools do we need to express those needs?
- How can we be sure there is a match between customer needs and the needs of the professionals and the solutions that can be found during the design phase? What references should be used for these evaluations?

There is a design-aid method available today that serves as a reference internationally: systematic design (Pahl and Beitz 1996). Initially (section 2), we will analyse how this method takes into account the notions of need. Second, we will examine the functional analysis method that was specifically developed to allow needs to be taken into account during the design phase. We will then (section 3) present our interpretation of that method and how it serves to answer the questions we are asking. We will put forward a design activities model, integrating customer needs and professionals' needs in the design phase on the basis of our interpretation. Our observations relative to the implementation of the functional analysis method (sections 5 and 6) will lead us to supplement this initial model and suggest tools that will allow this new federating model to function.

2. The expression of the needs of customers and professionals in the systematic design method

Numerous design methods exist and have been referenced by several authors (Finger and Dixon 1989, Suh 1990). The systematic design approach (Pahl and Beitz 1996), first put forward in 1970 by Professors Pahl and Beitz, is now recognized world-wide. Having detailed the different phases of the methodology and analysed them in the context of our own postulates, we will show the extent to which the different works currently underway are tending to supplement the method. The development of the organisation of project groups' activities is taken into account as part of a move towards integrated design methodology.

2.1. The systematic design method

Pahl and Beitz postulate that the process for finding a solution to a design problem is identical to that used for solving problems of a more general nature. When the designer designs, he follows an approach that comprises certain fundamental activities: formulation of needs and problems, the search for alternative solutions, evaluation, documentation and communication of results. Pahl and Beitz divide the design activity into four distinct phases (figure 1), which are organized according to a process that is, *a priori*, sequential.

Phase 1: planning and clarifying the task. An information search occurs during this phase to identify the needs the product will be required to meet (market, company, economy), the constraints that have to be taken into account (manufacturing, assembly, maintenance, recycling, etc.) and an initial project outline. This phase produces an initial design specification.

Phase 2: conceptual design. This phase begins with an analysis of the aforementioned specifications, with a view to identifying the major problems that need to be resolved. These problems are then formulated in a way that is supposed to be neutral, which means that no solution is pre-supposed. Reflection in a broader sense can then begin as to the possible solutions available. The problem is at this point divided up into subproblems, and function groupings are studied. This phase requires creative methodology and conventional or systematic research methods (patents, etc.). The most promising solutions are then developed and expanded upon in the form of concepts (solution principles).

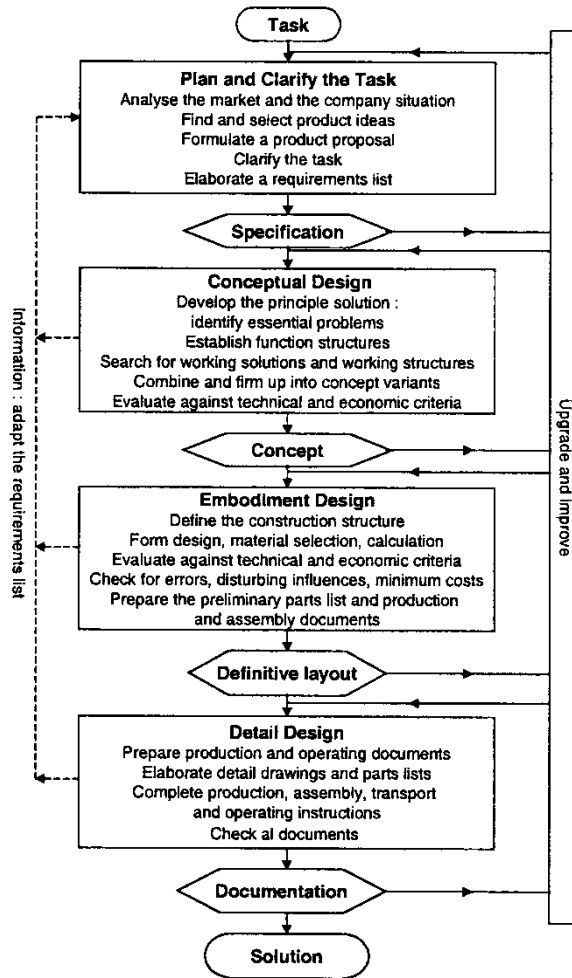


Figure 1. The design process according to Pahl and Beitz.

Phase 3: embodiment design. During this phase, the designer takes the concepts, determines the structure of the technical system and creates the forms of the final concept. In this phase, numerous alternatives have to be taken into account from a product point of view. The system will be developed up to such a time as checks can be made to ensure it performs the required functionalities, to verify its sustainability, production feasibility, aspects associated with assembly and so forth. Pahl and Beitz provide guidelines constructed upon rules and principles to fulfil these activities.

Phase 4: detail design. In the detailed study, the final drawings and documents are drawn up. Most high-level decisions have already been taken prior to this project phase. Nevertheless, care must be taken during the optimization phase, as solutions can still be very much open to discussion.

Applying this approach, we note that the expression of customer needs and those of the professionals are taken into account at two different levels:

- For buyers/end-users/customers, this is the first phase (clarifying the task), which refers back to the customer need. The tools used are market and company environment analysis tools. However, it is not clear how this expression is subsequently taken into account. The customer need is considered in a global sense without any refined characterization, which would have allowed the product to be assessed during the design phase.
- The needs of the professionals appear in this phase in the form of generic constraints, which are also difficult to take into account since they are not very clearly described at this point. A checklist of possible constraints that need to be taken into account is suggested in order to systematically review each stage of the product life-cycle: assembly, maintenance, recycling, and so on. This same checklist is used in the conceptual design phase, in an attempt to take account of these constraints during the preparation of solutions. In fact, the needs of the downstream actors appear more specifically at the embodiment design phase (i.e. once the physical principles of the solution have been established and a number of concepts put forward). A certain number of recommendations come out of the 'Design for X' method.

We also note that, in their method, Pahl and Beitz describe links between the different phases to ensure coherence between the different levels of the system under investigation, but no tools or methods are suggested to ensure information is exchanged. Indeed, the tools put forward—recommendations, guidelines and rules—allow design problems specific to each phase to be resolved, without creating any real link between them. Thus, the highly linear aspect of the development of the methodology requires a highly sequential and prescriptive design between the different phases and therefore between the different actors involved.

On the basis of these observations, several authors have proposed methods or tools that bring more substance to their ideas to enhance the integration of the different needs and improve solution evaluation during the solution-identification process itself.

2.2. Developments that supplement the method

In the realms of artificial intelligence, specialists have attempted to put in place computer tools to aid the design function. The fundamental hypothesis for most of them is that there is a nucleus of reasoning and types of knowledge that are common to the different domains involved in the design activity. Iwaski and Colleagues (see Vescovi et al. 1993) attempted, for example, to create a language that would allow them to specify the functions required of a product. This language enables us to both explicitly situate the physical context in which the function has to operate and to describe the function as the expected cause of a sequence of events. Through this language, they seek to give explicit representations of the function–physical principles interaction and of the constraints associated with such interactions.

Deng et al. (2000a) demonstrate that the functional verification aspects of design are very often neglected by the research community. To solve this problem, they have set up a 'complete' functional design model. They propose strategies to reduce the number of design variables, highlight dependencies, propagate constraints and operate

dynamic design checks. A design model was created for routine design activities (Deng et al. 2000b).

Taylor and Henderson (1994) looked at the role of features in the design of mechanical systems, as did Brunetti and Golob (2000), who proposed a feature-based approach to the creation of a product model. This model allows information that is specific to the different phases of the method, from the conceptual design phase to the detail design phase, to be expressed. The feature-based structure means that information on such things as functions, operating principles, physical effects and solution principles can be manipulated. The level at which the solution principle is examined is then matched up with the level where the functions are represented, itself being associated with the product level (the solution). By extracting the semantics of the product as early as the conceptual design phase, they can be used throughout the whole of the project.

Malmqvist (1995) proposed initially to match up a product model and the function/mean tree so as to build up a product design history. Subsequently, this model was extended to the different levels of design (Malmqvist and Schachinger 1997) and the systematic design approach broadened by using some of the components of the TRIZ method (Malmqvist et al. 1996).

We can observe that these different approaches are mainly intended to consider the coherence of models used for product representation through the different phases of the Pahl and Beitz method. Even though the models mean that coherence can be partially assured for the developments of the different phases, we can nevertheless contend that:

- None of the models provides a solution to the problem of characterising functions and constraints during phase 1, 'planning and clarifying the task', and that the needs being considered are only those of the customer.
- That research concentrates primarily on the conceptual design phase with a view to enhancing the links with the embodiment design phase (the progression from embodiment design to detail design is frequently treated as a partially resolved problem). The functions referred to are mostly those associated with product components (technical functions) and not the functions required in response to customer needs. The main objective of the research, then, has been to define the technical functions required of the product and for them to be squared up to the possible physical principles leading to the definition of a solution (Schachinger and Johansson 2000).

On the basis of these observations, we propose (external or internal) functions, constraint and assessment criteria concepts as means of answering our two initial questions. These concepts are derived from the normalized European method called functional analysis. In the forthcoming sections, we shall present its concepts and methodology and put forward a model for the design activities it suggests.

3. The concepts and principles of functional analysis

Functional analysis is a normalized design method (see Normes NF X 50–100 1996, Normes NF X 50–150 1990, Normes NF X 50–150-1 1996, Normes NF X 50–151 1991). Initially (1985), it was seen as an initial stage of value analysis. Within this framework, the objective is to characterize the needs that have to be

satisfied for the end-user or the buyer, by contributing to a complete description of the functions and their relations. Those are systematically characterized, classified and assessed. A function is described as the action of a product or one of its constituent parts. It is expressed, only in terms of finality, by a verb followed by a complement (e.g. 'transform energy'). The functional specification becomes the contractual document to which the design office is required to adhere. However, it is impossible to take only the customer's needs into account as a way of providing the service required without taking into account the potential resources that exist to make that service available. Considering the objects he considers artificial, Simon (1969) states that:

An artefact can be thought of as a meeting point—an 'interface' in today's terms—between the 'inner' environment, the substance and organization of the artefact itself, and an 'outer' environment, the surroundings in which it operates. If the inner environment is appropriate to the 'outer' environment, or vice versa, the artefact will serve its intended purpose.

Since 1996, the method has used the concepts of external functional analysis and internal functional analysis, which are very close to those put forward by Simon:

External functional analysis (or the functional analysis of need), lists the services the product is required to provide irrespective of the means available to it to provide them. It can be carried out on the basis of the consideration of the different situations in the product life-cycle (use, manufacture, maintenance, recycling, etc.). The product is, at this point, considered a black box, which is able to satisfy customer needs and those of the professionals involved in making it. It is described in terms of service functions and constraints. The service functions define (figure 2): (1) the interaction functions (IF) that correspond to the services provided by the product during the product life-cycle; and (2) the adaptation functions (AF) that reflect reactions, resistance or adaptations to elements found in the outside environment (E1, E2, E3). These functions are obtained from systematic analysis of the possible relations between: (1) the different elements of the outside environment via the product for IF; and (2) the elements of the outside environment and the product being designed for AF. Different tools are put forward (e.g. graph of interactors; figure 2) to allow these needs to be expressed.

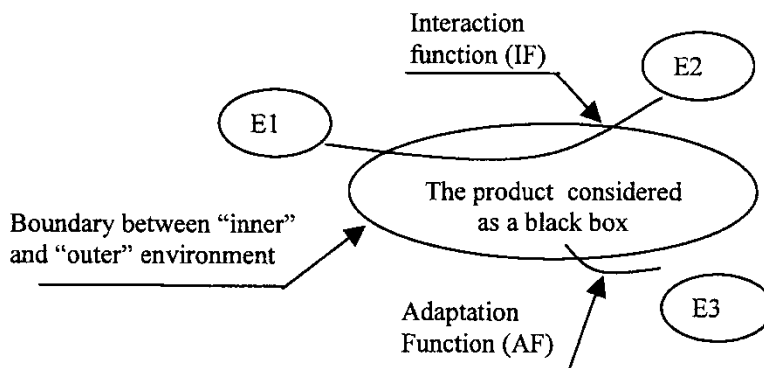


Figure 2. External functional analysis and graph of interactors.

A constraint is defined as a design characteristic, effect or provision for design, which is made compulsory or forbidden for whatever reason. A note specifies that two types of constraints have to be taken into account: those concerning solutions and those concerning the aim (in this case, the functions).

Our interpretation is as follows:

- The constraints that concern the aim characterize the relationship the product has with its environment: these are limitations on the product that are demanded either by the customer (primarily defined by the performances required) or by the market (referring to economic registers by cost, commercial registers by notions of deadlines or quality, and legal registers through laws and norms, company strategies and so forth).
- Those constraints that concern solutions are limitations imposed either by the technological solutions possible (primarily provided by the technical characteristics of the components), or by the company's technical strategies (potential production or maintenance facilities, type of maintenance planned: maintainable or disposable, preferences of the buying department, etc.).

These constraints can impact either in the same form as the functions (e.g. be maintainable) or through specifications that characterize the performances expected (e.g. a lifespan of x number of operating hours).

Internal functional analysis (or technical functional analysis) enables us to analyse the resources required and the way they are allocated to provide the service required. It is carried out from an internal point of view—from the standpoint of the designer(s) responsible for providing the system that will cover the needs identified. One of the key concepts of this type of analysis is that of the technical function, defined as 'the action of a constituent part or an action between the constituent parts of the product designed to provide the service functions required'. The FAST is an example of a tool made available to identify the technical functions that could potentially fulfil the service functions, and then to match up those technical functions with potential solutions.

The functional expression of needs does not preclude the search for technological principles, physical principles and possible solutions for making them happen. However, this search does come after the expression in functional terms (internally and externally). This manner of organising activities is presented as a means of promoting innovation and bringing to the fore the purposes of the product rather than the means for arriving at those purposes.

Functional analysis is therefore intended as a structured process (figure 3), 'moving progressively from the need to how it is satisfied, from the result to be obtained to the elements to be put in place to achieve those results'. With the design activities being organized sequentially in this way, there is a big risk that the customer's needs will take priority, and that they will rule the activity of all those involved in design and implementation. The method does, however, provide various opportunities for taking a step backwards (see figure 3) in order to take into account the needs of the professionals involved in the expression of the functions of service, after the initial stages have been gone through. The first way back is specified after a technological feasibility study of the problem exposed through the functional expression of the need (the search for technological solutions and principles). The second is realized further to the

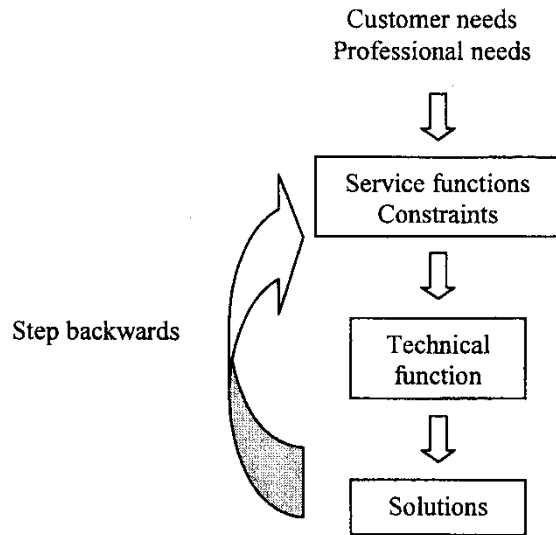


Figure 3. Structured and sequential process induced from functional analysis.

completion of a pre-project study (the search for a solution to a chosen principle), presenting one or more technological responses to the needs thus expressed.

The concepts of service function and constraints, as interesting as they may be, are not sufficient to characterize the need. Functional analysis proposes to add on the concepts of assessment, level and flexibility criteria. The assessment criterion is defined as 'a characteristic that is used to evaluate the performance expected of a product'. The level is the position on 'the scale of measurement of the assessment criterion'. The flexibility is 'an indication expressed by the buyer as to the possibilities of modulating the level required for a given assessment criterion'. This shows what is explicitly considered as being negotiable in the formulation of a problem. Therefore, the act of characterizing a service function or a constraint involves identifying all the assessment criteria, their level and flexibility, to evaluate whether or not the function has been achieved during the design phase. This leads us on to putting more meat on the functional analysis model, by adding assessment criteria to service functions and constraints (figure 4).

As we outlined earlier, the functional analysis method considers the expression of the needs of the various actors of the product life-cycle as essential. This method proposes concepts (functions, constraints, assessment criteria) and tools (Graph of Interactors, FAST, etc.) to make them work and to construct relations between needs and constraints, service functions, technical functions and solutions, stage by stage. Functional analysis can thus be considered a guideline for the whole design operation, as a dynamic way of looking at a problem to which all designers involved attempt to respond in a collaborative manner. Indeed, the method is intended to be a pluridisciplinary one, bringing together the various actors of the product life-cycle when these needs are expressed. The fact remains, however, that the model is highly sequential and that, while the assessment criteria can be used as references during evaluations, these are external criteria. What, then, are the links with those mobilized internally in making the choice between the different solutions?

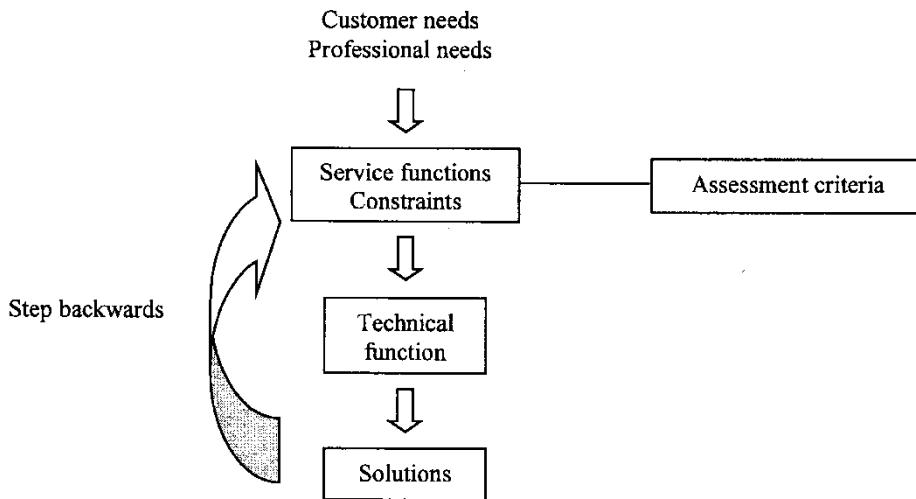


Figure 4. Model concerning design activities induced from functional analysis.

We have observed different situations for the implementation of this method in industrial or training contexts. In section 4, on the basis of a representative example of our observations, we will show how we can add to our model, taking into account the fact that functions/constraints and solutions are defined jointly through the whole of the design process. We will also demonstrate the important role played by certain parameters characterizing existing solutions that are mobilized in the assessment actions and in the organization of design activities. Using another example, we will refine this model still further in section 5, while taking into account the characterization of technical functions.

4. Taking the need into account: functions and solutions being defined simultaneously

4.1. Description of the development of a project

Here, we are going to outline our observations on a design process that uses the functional analysis method. The example used is the development of a throttle for regulating the speed of electric locomotives. At the outset, a pluri-disciplinary work group had set the target of thinking about the design of a future Train à Grande Vitesse (TGV) driver's cabin. The group comprised the designers from the design office, ergonomics engineers operators and drivers from the various regions of France.

The study began with a detailed analysis of the driving activities required in existing TGVs along with the collection of a great deal of data relating to train driving: the number of operations using the controls, manipulation times, driving strategies, and so on. Users' strict speed regulation needs were sought and determined in the course of discussions with mechanics and during observations of existing systems. These needs were identified while taking into account the constraints of the market, given the impact of the system on passenger transport (timetables, climate conditions, condition of the equipment, passenger comfort, etc.) and from the expression of the objectives required of the control itself (precise regulation, speed, safety, etc.).



Figure 5. 'TGV Atlantic' driver's cabin.

This first stage provided a good understanding of the TGV train-driver's job, and of the mechanisms currently in use for speed regulation (figures 5 and 6): the pneumatic brake control, the circular electrical traction/braking control and the push-button controls for emergency stops. This analysis was completed with an information search on existing mechanisms.

During the project group discussions, a large amount of information came to light on the future developments of the train-driver's job, the current expectations of drivers, mechanisms under development, and so on. Thus, the members of the project group were able to progressively construct a common representation of the functions that would need to be operated by the new mechanism and the modifications that would be required for the proper functioning of the main speed controller.

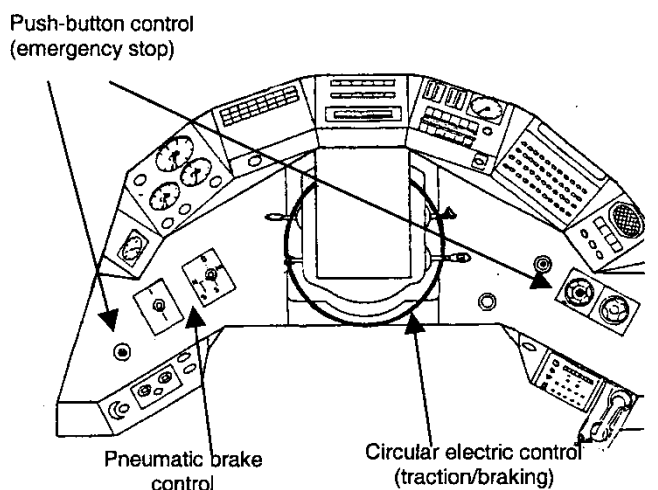


Figure 6. TGV's traction/braking controls.

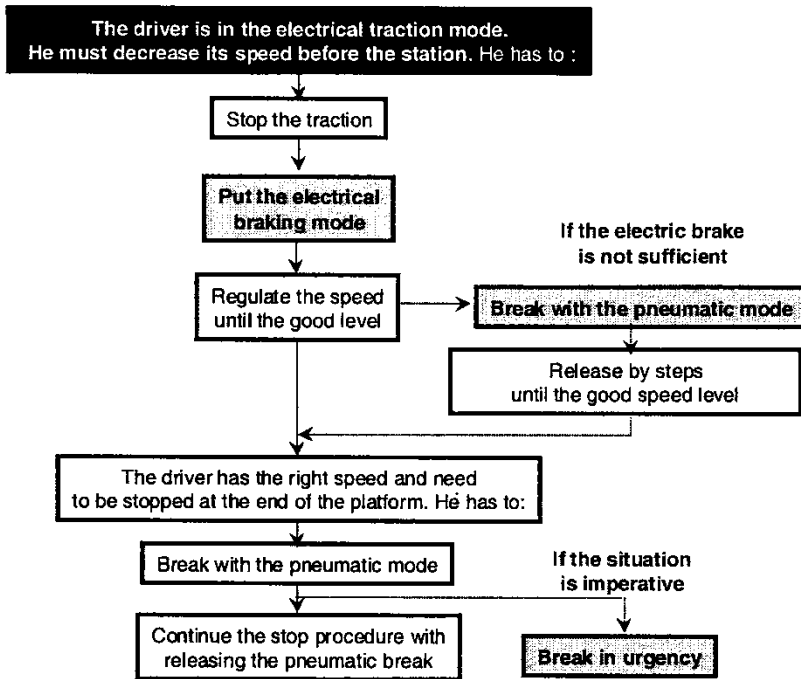


Figure 7. Scenario to illustrate the arrival to the station.

The development of scenarios linked to the use of the control (figure 7) allowed the functional expression of the need to be refined as part of the development of the future speed control throttle.

By way of an example, we have presented an extract of the recap of future constraints to be respected (Table 1) once the practical scenarios had been considered, which contributed to the preparation of the functional specifications of the future mechanisms.

Proposals for technological solution principles were then gathered, inspired in particular by existing mechanisms, especially from the aeronautics field. The evaluations carried out led the project group to abandoning the principle of having a rotating joystick control (a choice that was nevertheless explicit in the constraints already listed), which would have produced too great a divergence from the level of precision desired for speed control. The principle of an impulse throttle was therefore considered, with the length of pulsations serving to regulate the levels of traction or braking required. The chosen throttle combined the three functions linked to the regulation of speed (figure 8): management of the electric traction and braking strain (+ or –), pneumatic strain management (Frein Conduite Générale—general driving brake) and emergency braking.

The modifications thereby made represent substantial changes to the mechanisms currently in use (switching from a circular control to an impulse control with visual display of levels on a screen, etc.). Clarification exercises had then to be undertaken to establish acceptable levels for assessment criteria such as the position of the control in relation to the driver and the intensity of the manipulation effort. On this specific

Concerned device	Analysis results	Future constraints to be respected
Traction/braking controls	• There are controls for each mode concerning traction/braking. This must be reconsidered • The actual circular control depend on an old technology (to manipulate a rheostat) • Procedures to change mode are complex and need a lot of actions. An action is necessary to release the actual mode and drivers have to put the level 0 • The control is adapted to drive with the left or right hand • The present control concerning pneumatic brakes give satisfaction to everyone	• The same control must integrate all modes concerning traction/braking • The throttle will be a rotating joystick control (small dimensions) • The action must be direct to change the mode without any action on another control and whatever the initial mode. The initialization (level 0), if necessary, will no imply another action • Efforts on the control must be adapted and will be done with the right or left hand • The principle of an impulse throttle for the pneumatic mode must be retained

Table 1. Extract of the recap of future constraints to be respected.

example of the quantification of manipulation effort, the project group chose to adopt a trial-based approach. It subcontracted the construction of a prototype, allowing several drivers to define the level of intensity of the effort required to operate the new control, in relation, of course, with the task of driving (photograph in figure 8). It is a fact that while this sort of information is readily available in the aeronautic sector, such data is difficult to apply to the rail sector, where the environment is very different.

The testers were thereby able to vary a number of technical parameters within the system (position of the links, stiffness of springs) and check at each stage if the control corresponded to the needs of the drivers (efforts required, repeatability, etc.). These simulations on prototype models meant that acceptable levels could be defined for several assessment criteria of the service functions, thus enhancing the specification previously drawn up, while taking the technical feasibility of the system into account.

4.2. Analysis

In this project, the search for service functions first led the members of the project group to carry out a full audit of what already existed, taking into account the driving equipment and the different applications; more than 10 drivers were observed actually driving. Further to this critical analysis of what already existed and taking into account the new needs for TGV throttles, an initial general expression of functions and constraints was formulated.

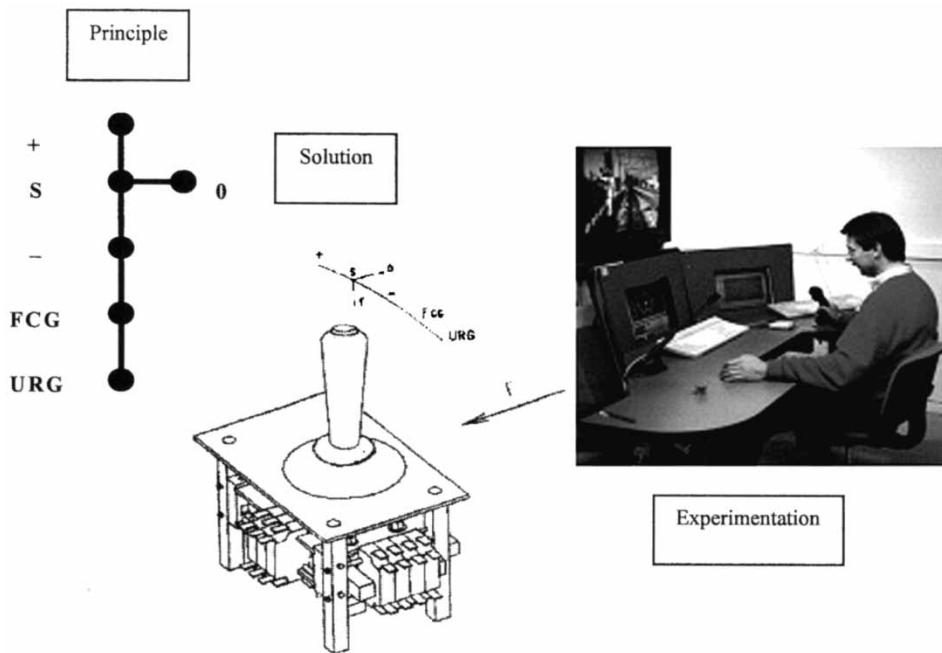


Figure 8. Simulations to establish acceptable levels of assessment criteria. FCG, general driving brake; URG, emergency braking.

Subsequently, discussions around the potential scenarios for the use of the equipment caused new constraints to emerge, and generated research into how a new technological principle could be created (joystick). Since the search for and evaluation of solutions to technological principles satisfied this new functional expression of the need, the group turned its attentions to controls that already existed in other areas of the 'driving' profession (aeronautics). The actors were therefore led to reconstruct the problem on the basis of a proposed alternative for the formulation of the specification, making the impulsion control a possible solution. But the change in the area of its use, from aeronautics to train driving, meant that the impulsion control had to be adapted to. So it was the prototype model, materializing the solution principle considered as a possibility, which allowed the technical principle to be validated, and the functions required to be clarified and specified.

This example shows us that the functional expression of the need was developed in an interactive manner with the construction of the need and the consideration of possible solutions (figure 9). Furthermore, the functional expression is not independent of the solution since it integrates the existence of technical parameters into the expression of external assessment criteria and associated levels specific to the solution. It should be noted that the scenario tools and the prototype model were constructed and implemented in the action, in such a way as to allow the different actors to have a grasp of the problem during its formalization. Implementation scenarios and the prototype model were used as tools to outline the functions and assessment criteria, respectively, for the former, and the assessment criteria in relation to the parameters of the solution for the latter. It should also be noted that the notion of technical function is absent from

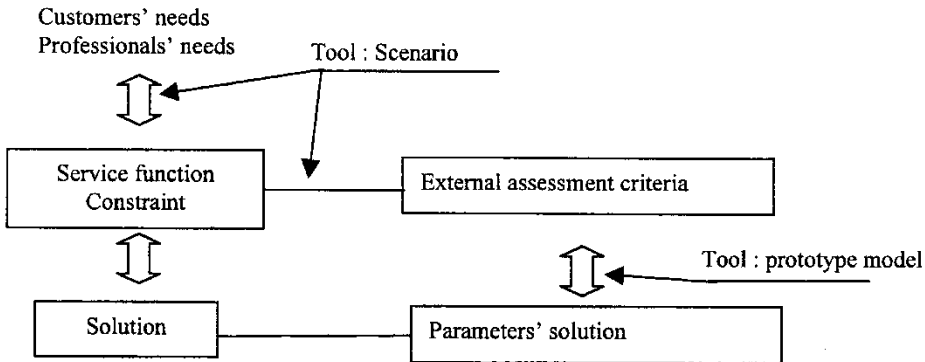


Figure 9. Links between needs, functions and solutions.

this model. This is certainly due to the fact that the prototype model was subcontracted. But the technical functions are nevertheless subjacent to the moment of the search for potential solution principles. They simply were not expressed as such.

Functions and solutions can therefore be considered as two aspects of the same product. The alternation between external functional analysis, the search for solution principles and solutions is clearly an opportunity for functional expression to express the needs of the different actors of the product life-cycle in a dynamic way. External functional expression also includes the description of assessment criteria that we describe as external, and which are to be linked to the technical parameters of possible solutions. The evaluation of solutions (positive or negative) on these parameters and their levels must lead to the same type of evaluation from an external point of view.

5. Product evaluation: the assessment criteria of functions through to solution parameters

The study we describe here concerns a ski resort cabin bay and, more specifically, a part whose service function is to brake the forward movement of the ski cabins when passengers are set down or picked up. The mechanical system under investigation comprises two pulleys mounted on their shafts (shafts 1 and 2) and a trapezoidal drive belt. At the arrival bay, it is the end-to-end assembly of 100 such systems that allows the braking and forward movement of the cabin to be controlled. Currently, the pulleys are made of aluminium and have the disadvantage of being heavy to handle during assembly and maintenance. The objective of the study was to identify whether, at minimum cost, it would be possible to change the material used to make the pulleys without changing the function of the mechanism.

An initial analysis brought out the technical functions that had to be respected in each of these systems so that the assembly performed the following functions:

- FT1: transmit power from shaft 1 to shaft 2,
- FT2: adapt the rotation speeds between shafts 1 and 2,
- FT3: be easily assembled and maintained, and
- FT4: resist outside weather conditions;

and the constraints having to be respected:

- Identical size to the present solution.
- Minimum cost.

The characterization of the functions led them to formulate assessment criteria and associate them with levels:

- For FT1: transmitted power >8 kW, transmission torque <320 Nm, operating life cycle = 24 000 hours, performance >0.96 .
- The size constraint was reflected in assessment criteria for FT1: diameter and width of pulleys, centre-to-centre distance between shafts 1 and 2.
- For FT3: weight of the pulley.
- For FT4: chemical resistance.
- The price constraint was reflected in the form of function FT5 whose price assessment criteria had to work to a minimum level.

Using the existing solution as the starting point, the designers formulated parameters defining it: materials, pulley groove primitive radius and angle; centre-to-centre, drive-belt length and wrapping angle, preload, means of attainment. Subsequently, they sought to determine which criteria would be affected by a variation in parameters.

This led to the construction of a so-called 'influence' matrix which showed the influence of each system parameter on the assessment criteria of the shaft–pulley–drive belt assembly (Table 2). Each shaded box in the table indicates that the parameter in question has a significant influence on the corresponding criterion and that it is that particular criterion that has to be acted upon to optimize the solution.

The influence of each parameter on the response to an expressed need was subject to a specific study, linked to calculation models. In figure 10, for example, we see the calculation model concerning the transmission torque, with the solution chosen for the torque transmission being a conical sleeve connection.

The primary function of table 2 was to modify the mental representations of the designers on the links between the parameters and the criteria. Then, a certain number of parameters were defined as being fixed by the project group. Only the parameters concerning the material to be used for the pulleys and the means of attainment were considered modifiable. During resolution, four types of material were considered (aluminium, cast iron, steel and polyamide). For each of these materials, the work involved checking whether the different assessment criteria could be respected. The evaluation of the polyamide solution was positive in relation to all the technical assessment criteria, with the exception of life expectancy. Indeed, the solution whereby there would be a complete link between the pulley and the drive shaft using the conical connection sleeve and a nut did not provide sufficient axial positioning precision. This lack of precision depend on the hardening of the polyamide when coming into contact with the drive shaft, and the level of imprecision on the tightening torque that had to be precise to ensure the right pressure stress. According to the manufacturer, this lack of precision would reduce the working life of the pulle/drive-belt assembly. This meant that a parameter had not been expressed in the description of the existing solution, only to appear during the search for potential solutions: the axial position of the pulley on the shaft. Several solutions were subsequently envisaged, keeping the polyamide material for the pulley: a flat prop with a cylinder, addition of a metal insert, combining a conical prop and a flat prop positioned in such a way as to allow hardening (and thus improve the torque transmission) and a precise axial position. This was the last solution to be chosen, since it meant the weight aspect and the cost could be optimized,

Function	Assessment criteria	Level	Pulley material	Primitive radius	Groove angle	Wrapping angle	Centre to centre distance	Belt length	Initial strain	Means of attainment
FT1	Transmitted power	>8 kW								
	Transmitted torque	>320 Nm								
	Operating life-cycle	24 000 hours								
	Performance	>0.96								
	Dimensions									
	Diameter	<400 mm								
	Width	<70 mm								
	Centre to centre distance	=500 mm								
FT2	Report de réduction	~1								
FT3	Weight of pulley	Minimum								
FT4	Chemical resistance	Atmosphere								
FT5	Cost	Minimum								

Table 2. An influence matrix to link external assessment criteria to solution's parameters.

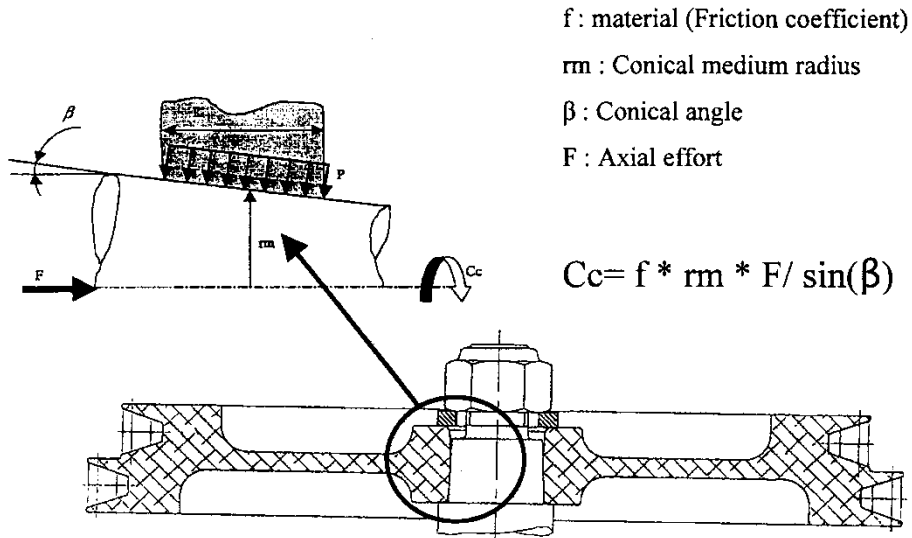


Figure 10. Torque calculation model for a conical sleeve connection.

meaning that it could be assessed positively in relation to all the assessment criteria characterizing the functions.

We envisage further development of the information contained within this matrix to allow us to specify the relative importance and the nature of the influence of the variations of parameters on assessment criteria. For example, is the transmission torque more dependent upon the material parameter than on the pulley radius parameter? Another example: if the radius is increased, the transmission torque will increase. But at the same time, the centrifugal effect on the belt will increase, the contact pressure will decrease and the transmission torque decline too. Which will be the most influential effect?

This example allows us to enhance the model proposed earlier by considering the following (figure 11):

- On the one hand, that assessment criteria can be combined with technical functions. This will enable us to talk about internal assessment criteria to differentiate from assessment criteria, which will be described as external, associated with service functions and constraints.

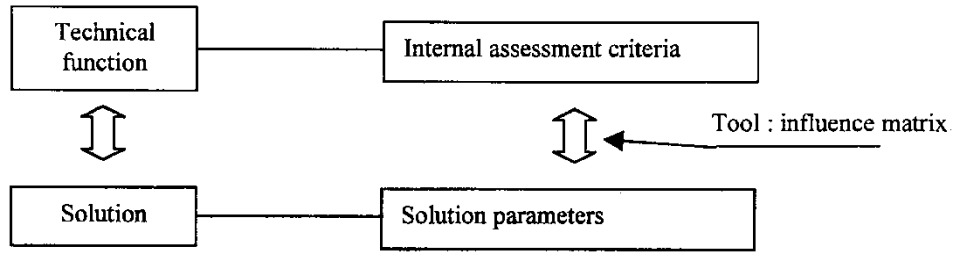


Figure 11. Links between solution Parameters and internal assessment criteria.

- On the other, that when we evaluate the solution, it will involve the parameters characterizing it, which are not assessment criteria but which have to be linked up to internal assessment criteria associated with technical functions. The influence matrix is a tool which allows us to construct this link.

The example outlined presents a description of technical functions that are very close to the solution. FAST shows that there are different levels of description of technical functions. If we consider that it is possible to associate internal assessment criteria with each level of description, we have to plan to have tools to establish the connection with the external assessment criteria characterizing the needs defined externally on the product.

6. Conclusion and future perspectives

The design activity is distributed (Brissaud and Garro 1996) in particular among the different actors involved in the product life-cycle. The integration of those actors from the earliest stages of design is explicitly intended as a means of allowing account to be taken simultaneously of their needs regarding the product under design. This led us to ask questions: What is the importance attributed to these needs by existing design methodology and what are the tools available to express them? What is the balance that existed between the solutions developed and the needs expressed?

To achieve this, we first of all looked at the systematic design method and the works carried out according to its principles in relation to the questions we were asking ourselves. At that point, we observed that the search for coherence between the different representations of the product concerns essentially the conceptual design, embodiment design and detail design phases, but that no link exists with the first 'planning and clarifying the task' phase during which the customer's needs are supposed to be expressed.

We then exposed the concepts and principles of functional analysis as developed in line with European norms. From that, we derived a methodological model, on the basis of the organization of design activities proposed therein. The observation of implementation situations using this method and its tools led us to the following conclusions.

- First, that the method does allow us to integrate the actors of the product life-cycle from the earliest stages of the design process. The tools it provides, and others put in place for the action (scenario, prototype model) that we suggest including in the method, led to the expression of the needs of the different entities involved and a translation of them into the form of functions, constraints and assessment criteria. The transfer to the different levels of these functions (service functions, technical functions) and constraints allowed us to create a link between the needs expressed and the solutions envisaged. The association of both assessment criteria with the functions and constraints (external or internal) and technical parameters to the solutions must allow us to create links between the reference elements for the evaluation of the product.
- Second, that the use of the functional analysis method leads to co-operation between the actors and to an organization of the different design activities that is quite different to that proposed in the method, itself very structured and linear. The example of the throttle for the TGV showed us that the co-ordination of the

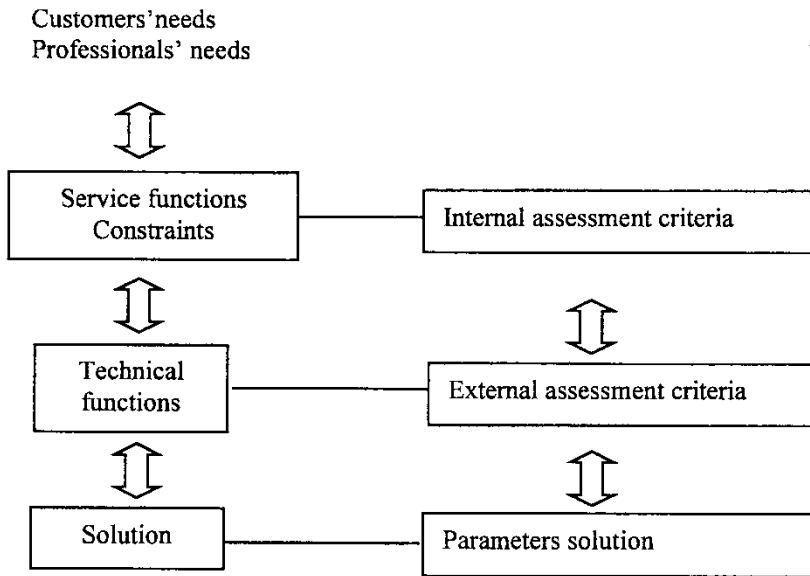


Figure 12. Design activities organization model proposed.

different activities is constructed in the action, relying on the simultaneous search for an explanation of the problem (service functions, constraints and external assessment criteria) and technically possible solutions. The problem and the possible solutions are constructed simultaneously.

We propose, therefore (figure 12), an organization model for these activities as described in section 4, integrating the model developed in section 5.

Functional analysis can therefore be considered a method that provides concepts and tools to manipulate the needs of those involved in the product life-cycle in a dynamic way, and this throughout the design process. Implementation scenarios, real or virtual prototype models and influence matrices complete the range of tools available and necessary for the expression of these needs and constraints and their matching up at different levels of granularity. It aims to construct coherent references for evaluations at those different levels. There is also the fact that the technical functions can also be broken down into different levels, and that we need to ask ourselves questions about the tools that will have to be developed to ensure the requirements drawn up can be properly tracked. This means creating links between the assessment criteria that characterize them at these different levels. Also, we feel that it will be necessary to create ways of identifying, memorizing and taking into account in the definition of the problem the assessment elements (function or criterion) that emerge during the action.

References

- BRISAUD, D. and GARRO, O., 1996, An approach to concurrent engineering using distributed design methodology, *Concurrent Engineering: Research and Applications*, 4, 303–311.
- BOOTHROYD, G., DEWHURST, P. and KNIGHT, W., 1994, *Product Design for Manufacture and Disassembly* (New York: Marcel Dekker).

- BRUNETTI, G. and GOLOB, B., 2000, A feature-based approach towards an integrated product model including conceptual design information, *Computer Aided Design*, 32, 877–887.
- DENG, Y. S., BRITON, G. A. and TOR, S. B., 2000a, Constraint-based functional design verification for conceptual design, *Computer Aided Design*, 32, 889–899.
- DENG, Y. M., TOR, S. B. and BRITTON, G. A., 2000b, A dual stage functional modelling framework with multi-level design knowledge for conceptual mechanical design, *Journal of Engineering Design*, 11, 347–375.
- DESCOTTE, Y. and LATOMBE, J. C., 1981, GARI: a problem solver that plans how to machine mechanical parts, *Proceedings of 7th IJCAI*, Vancouver (San Francisco: Morgan Kaufmann Publishers), pp 766–772.
- FINGER, S. and DIXON, J. R., 1989, A review of research in mechanical engineering design, *Research in Engineering Design*, 1, 51–67.
- GARRO, O., BRISSAUD, D. and TICHKIEWITCH, S., 1996, Toward a cooperative design system, *COOP 96*, Juan-les-Pins, France (Rocquencoust: COOP Group), pp 151–162.
- MALMQVIST, J., 1995, A computer-based approach towards including design history information in product models and function-means trees, *Proceedings of the Design Theory and Methodology Conference, DTM'95*, Boston, MA (Boston: ASME publication), pp 593–602.
- MALMQVIST, J. and SCHACHINGER, P., 1997, Towards an implementation of the chromosome model focusing the design specification, *IECD'97*, Tampere, Finland (Tampere: Tampere University of Technology), pp 203–212.
- MALMQVIST, J., AXELSSON, R. and JOHANSSON, M., 1996, A comparative analysis of the theory of inventive problem solving and the systematic approach of Pahl and Beitz, *Proceedings of the 1996 ASME Design Engineering Technical Conference*, Irvine, CA (Irvine: ASME publication), paper number 96-DETC/DTM-1529.
- MAFFIN, D., 1998, Engineering design models: context, theory and practice, *Journal of Engineering Design*, 9, 315–327.
- NORMES, NF X 50–100, 1996, Analyse fonctionnelle, Caractéristiques fondamentales (Paris: AFNOR).
- NORMES, NF X 50–150, 1990, Analyse de la valeur, Analyse fonctionnelle, Vocabulaire (Paris: AFNOR).
- NORMES, NF X 50–150-1, 1996, Vocabulaire du management, de la valeur, de l'analyse de la valeur et de l'analyse fonctionnelle (Paris: AFNOR).
- NORMES NF X 50–151, 1991, Expression fonctionnelle du besoin et cahier des charges fonctionnel (Paris: AFNOR).
- PAHL, G. and BEITZ, W., 1996, *Engineering Design*, 2nd edition (New York: Springer-Verlag).
- PAWAR, K. S., HAQUE, B. and BRASON, R. J., 1999, Establishing concordance within engineering teams, *Concurrent Engineering: Research and Applications*, 7, 215–229.
- SCHACHINGER, P. and JOHANESSON, H., 2000, Computer modeling of design specifications, *Journal of Engineering Design*, 11, 317–329.
- SIMON, H. A., 1969, *The Science of the Artificial* (Cambridge: MIT Press).
- SOLHENIUS, G., 1992, Concurrent engineering, *Annals of the CIRP*, 41, 645–655.
- SUH, N. P., 1990, *The Principles of Design* (New York: Oxford University Press).
- TAYLOR, L. E. and HENDERSON, M. R., 1994, The role of feature and abstraction in mechanical design, *Proceedings of the 6th Conference on Design Theory and Methodology, DTM'94* (Minneapolis, MN: ASME), pp. 131–140.
- VESCOVI, M., IWASAKI, Y., FIKES, R. and CHANDRASEKARAN, B., 1993, CFRL: a language for specifying the causal functionality of engineered devices, *Proceedings of the 11th National Conference on AI (AAAI-93)* (Cambridge, USA: AAAI Press/MIT Press), pp. 626–633.

